

Shubham Bhattacharya

Software Engineer — GenAI / LLM Applications (TypeScript)

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Available: Internship now · Full-time Dec 2026 · Open to relocate / remote

SKILLS

Languages: TypeScript, JavaScript (Node.js, Bun), SQL
Backend: Express.js, Fastify, REST API Design, SSE Streaming
Databases: PostgreSQL, pgvector, Prisma ORM, Redis
AI & LLMs: RAG (hybrid retrieval, chunking, cosine-gated pgvector), Agents (custom ReAct, Zod-validated tool registry), Evals (golden sets, grounding/attribution tests), Multi-provider failover (Gemini/Groq/OpenRouter), Streaming (SSE), Prompt-injection mitigation
DevOps: GitHub Actions CI/CD, Docker Compose, Vercel, Render, Sentry, Pino Logging
Fundamentals: Data Structures & Algorithms, DBMS, OS, Networks (BCA coursework)

PROJECTS

RAG Chat — Grounded Document Intelligence (*Bun, Fastify, TypeScript, pgvector, Gemini, Groq, SSE*) — Jul–Sep 2026 · Solo
github.com/shubhambhattacharya-dev/RagChatBot | [Live Demo](#)

- **Hybrid RAG retrieval** — metadata + lexical + cosine-gated pgvector (**distance 0.5**): weak evidence never reaches the LLM; the system **refuses instead of hallucinating**.
- **Evaluation harness** — 14/14 golden-set cases passed incl. an adversarial trap case (model knows, corpus doesn't); retrieval gated at cosine distance 0.5.
- **SSE streaming with async ingestion (BullMQ + Redis)** keeping 10–30s extraction off the request path.

AI Ops — AI Customer Support Backend (*Node.js, TypeScript, Express, Groq, PostgreSQL, Prisma, Zod, Sentry*) — May–Aug 2026 · Solo
github.com/shubhambhattacharya-dev/Agentic_Project | [Live Demo](#)

- **Custom ReAct agent** (no LangChain) — order-tracking + refund drafts via auditable multi-step reasoning.
- **Zod-validated tool registry** — every LLM response schema-checked before any DB write or API call: blocks malformed and hallucinated tool calls before any DB write.
- **LLM failover routing** (Llama 70B → 8B) — **1200ms measured recovery**.

Medo Copilot — AI Launch-Readiness Auditor (*Next.js, TypeScript, Vercel AI SDK, Clerk, Gemini, Groq, OpenRouter*) — Apr–Aug 2026 · Solo
github.com/shubhambhattacharya-dev/Medo_Copilot

- **3-tier LLM failover waterfall** (Gemini → Groq → OpenRouter) — automatic provider failover for high availability.
- **Hybrid scoring engine** — deterministic Lighthouse/static metrics blended with LLM analysis.

DocNow — Telemedicine SaaS Platform (*TypeScript, Prisma, PostgreSQL, Vonage WebRTC*) — Feb–Mar 2026 · Solo
github.com/shubhambhattacharya-dev/health | [Live Demo](#)

- **Symptom triage engine** routing patients to the right specialist based on reported symptoms.
- **Browser-based video consults** (Vonage WebRTC, no app download) + credit wallet with **full audit trails** for purchases and refunds.

EDUCATION

Bachelor of Computer Applications (BCA)

Vadodara, Gujarat

Parul University · Expected Dec 2026

CGPA: 8.13 | First Class with Distinction · Relevant: DBMS, OS, Networks, Software Engineering